

Exact Incompressible Smoothing for Unitriangular Deformations and Radial Twists

Candidate partial result for arXiv:2502.01336

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Status. Candidate partial result; likely valid; human review requested. The results below solve structured subcases of Open Problem 5 in Hencl’s survey, not the general incompressible Ball–Evans approximation problem.

1 Source question

The source is Stanislav Hencl, *Ball–Evans approximation problem: recent progress and open problems*, arXiv:2502.01336v1 (3 February 2025) [1]. Open Problem 5 appears on PDF page 6.

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incompressible diffeomorphisms and this could be important e.g. in problems related to [1]. In fact the following open problem was asked to me by Assis Azavedo.

Open problem 5. Ball–Evans mapping for incompressible homeomorphisms: Let $f \in W^{1,p}(\Omega, \mathbb{R}^n)$ be a homeomorphism with $J_f \equiv 1$ a.e. Can we find a sequence of incompressible diffeomorphisms (or piecewise affine incompressible homeomorphisms) which approximate f in the Sobolev norm? Note, that this is open already in the planar case $n = 2$.

Some models of Nonlinear Elasticity include in fact also the second gradient and we work with the functional

In transcription, the question asks whether a homeomorphism $f \in W^{1,p}(\Omega, \mathbb{R}^n)$ with $J_f = 1$ almost everywhere can be approximated in the Sobolev norm by incompressible diffeomorphisms (or by piecewise affine incompressible homeomorphisms). The survey emphasizes that the question is open even for $n = 2$.

2 Results and proof intuition

Here “incompressible” means that the Jacobian determinant is identically one (almost everywhere for a Sobolev map and pointwise for a smooth map).

Proof intuition

There are two independent mechanisms.

First, a unitriangular deformation changes coordinate i only by a function of the later coordinates. Its derivative is upper triangular with diagonal entries one, so ordinary smoothing of the scalar coordinate functions cannot change the determinant. The same order of the coordinates gives an explicit back-substitution inverse, so smoothing cannot destroy injectivity either.

Second, a radial twist rotates every sphere $\{|x| = r\}$ through an angle that depends on r . Smoothing the angle leaves the radius fixed. Thus every approximant maps the ball onto itself. In the derivative, the only extra term is rank one; the matrix determinant lemma reduces its contribution to a quadratic form of a skew-symmetric matrix, hence to zero. Freezing the angle near the origin costs arbitrarily little in the exact weighted Sobolev energy and makes the approximants smooth at the origin.

The first theorem covers a large triangular class and, in particular, one-dimensional rank-one laminates. The second theorem has the additional features that the source and target ball are fixed and the boundary trace is preserved. General incompressible homeomorphisms need not have either structure, and remain outside the argument.

3 Unitriangular deformations

For intervals $I_j = (a_j, b_j)$ put

$$Q = I_1 \times \cdots \times I_n, \quad Q_i = I_i \times \cdots \times I_n.$$

Lemma 1 (Simultaneous uniform and Sobolev smoothing). *Let $V \subset \mathbb{R}^d$ be a bounded open rectangle, $1 \leq p < \infty$, and $u \in W^{1,p}(V) \cap C(\overline{V})$. There are functions u_k , smooth on a neighborhood of $\overline{V}$, such that*

$$\|u_k - u\|_{W^{1,p}(V)} + \|u_k - u\|_{L^\infty(V)} \longrightarrow 0.$$

Proof. Reflect u successively across the faces of V . On a slightly larger rectangle this gives a continuous $W^{1,p}$ extension. Convolution with a standard mollifier converges in $W^{1,p}$ by the usual translation continuity argument, and converges uniformly because the continuous extension is uniformly continuous on a compact neighborhood of $\overline{V}$. $\square$

Theorem 2 (Exact smoothing of unitriangular deformations). *Let $n \geq 2$ and $1 \leq p < \infty$. For $i = 1, \dots, n-1$, suppose*

$$u_i \in W^{1,p}(Q_{i+1}) \cap C(\overline{Q}_{i+1}),$$

and let $c \in \mathbb{R}$. Define $F : \overline{Q} \rightarrow \mathbb{R}^n$ by

$$\begin{aligned} F_1(x) &= x_1 + u_1(x_2, \dots, x_n), \\ F_2(x) &= x_2 + u_2(x_3, \dots, x_n), \\ &\vdots \\ F_{n-1}(x) &= x_{n-1} + u_{n-1}(x_n), \\ F_n(x) &= x_n + c. \end{aligned}$$

Then $F|_Q$ is an incompressible $W^{1,p}$ homeomorphism of Q onto $F(Q)$. There are smooth incompressible diffeomorphisms $F_k : Q \rightarrow F_k(Q)$ of the same unitriangular form such that

$$\|F_k - F\|_{W^{1,p}(Q)} + \|F_k - F\|_{L^\infty(Q)} \longrightarrow 0.$$

The conclusion holds in every dimension and for every finite p , including $p = 1$.

Proof. Apply Lemma 1 to each u_i , obtaining smooth $u_{i,k}$, and define F_k by the displayed formula with u_i replaced by $u_{i,k}$. The derivative matrices DF almost everywhere and DF_k everywhere are upper triangular, with every diagonal entry equal to one. Therefore

$$J_F = 1 \quad \text{a.e.}, \quad J_{F_k} = 1 \quad \text{everywhere.}$$

Injectivity does not require an estimate. Given $y = F(x)$, first recover $x_n = y_n - c$, then

$$x_{n-1} = y_{n-1} - u_{n-1}(x_n),$$

and continue backwards until x_1 is recovered. These formulas give a continuous inverse on $F(Q)$. The same formulas, with smooth $u_{i,k}$, give a smooth inverse for F_k . They also show directly, by the openness of the interval constraints under the recursive recovery, that $F(Q)$ and $F_k(Q)$ are open. Thus F is a homeomorphism onto its image and every F_k is a diffeomorphism onto its image.

Let $\pi_i(x) = (x_{i+1}, \dots, x_n)$. Fubini's theorem gives, for either the function or any derivative in a tail variable,

$$\|(u_{i,k} - u_i) \circ \pi_i\|_{L^p(Q)}^p = \left(\prod_{j=1}^i |I_j| \right) \|u_{i,k} - u_i\|_{L^p(Q_{i+1})}^p.$$

The $W^{1,p}$ convergence follows after summing over i , while uniform convergence is immediate from the coordinate formula. $\square$

The following is the rank-one laminate form that motivated the construction.

Corollary 3 (One-dimensional incompressible laminates). *Let $U \subset \mathbb{R}^{n-1}$ be a bounded open rectangle and $I = (a, b)$. Let $B : \mathbb{R}^{n-1} \rightarrow \mathbb{R}^n$ be injective and let $\phi \in W^{1,p}(I; \mathbb{R}^n)$. Suppose*

$$\det[Be_1 \mid \dots \mid Be_{n-1} \mid \phi'(t)] = 1 \quad \text{for a.e. } t \in I.$$

Then

$$f(x', t) = Bx' + \phi(t)$$

is an incompressible homeomorphism onto its image and is the uniform and $W^{1,p}$ limit of smooth incompressible diffeomorphisms onto their images. The approximating curves can be chosen to have the same endpoint values as ϕ .

Proof. Let

$$\Delta(v) = \det[Be_1 \mid \dots \mid Be_{n-1} \mid v].$$

This is a nonzero linear functional, $\ker \Delta = \text{ran } B$, and the hypothesis says $\Delta(\phi') = 1$. Choose v_0 with $\Delta(v_0) = 1$. Absolute continuity on the interval gives

$$\phi(t) = \phi(a) + (t - a)v_0 + Bw(t)$$

for some $w \in W^{1,p}(I; \mathbb{R}^{n-1})$ with $w(a) = 0$. This is, after fixed affine changes of variables, a one-layer instance of Theorem 2.

For completeness, approximate w in $W^{1,p}$ by smooth curves and preserve both endpoint values by the affine correction

$$\tilde{w}_k(t) = w_k(t) + \frac{b-t}{b-a}(w(a) - w_k(a)) + \frac{t-a}{b-a}(w(b) - w_k(b)).$$

The trace map on a finite interval is continuous on $W^{1,p}$, so this correction tends to zero in $W^{1,p}$. It stays in $\mathbb{R}^{n-1}$, hence after multiplication by B it cannot change $\Delta(\phi'_k) = 1$. Finally,

$$\Delta(f(x', t) - \phi(a)) = t - a$$

recovers t from the image, after which injectivity of B recovers x' . $\square$

4 Radial twists of the ball

Let $\mathbb{B}^n = \{x \in \mathbb{R}^n : |x| < 1\}$. For $s \in \mathbb{R}$, let $R_s \in SO(n)$ be rotation through angle s in the (x_1, x_2) -plane and the identity on its orthogonal complement.

Theorem 4 (Fixed-ball smoothing of radial twists). *Let $n \geq 2$, $1 \leq p < \infty$, and let $\theta : (0, 1] \rightarrow \mathbb{R}$ be locally absolutely continuous with*

$$\int_0^1 r^{n-1+p} |\theta'(r)|^p dr < \infty. \quad (4.1)$$

Define

$$T_\theta(0) = 0, \quad T_\theta(x) = R_{\theta(|x|)}x \quad (x \neq 0).$$

Then T_θ is an incompressible $W^{1,p}$ homeomorphism of $\mathbb{B}^n$ onto itself. There exist $\theta_k \in C^\infty([0, 1])$, each constant near zero and satisfying $\theta_k(1) = \theta(1)$, such that

$$T_{\theta_k} \longrightarrow T_\theta \quad \text{in } W^{1,p}(\mathbb{B}^n; \mathbb{R}^n) \text{ and uniformly on } \overline{\mathbb{B}^n}.$$

Every T_{θ_k} is a smooth incompressible diffeomorphism of $\mathbb{B}^n$ onto itself, and $T_{\theta_k} = T_\theta$ on $\partial\mathbb{B}^n$. No limit of $\theta(r)$ as $r \downarrow 0$ is required.

Proof. Write $r = |x|$, $e_r = x/r$, $Q(r) = R_{\theta(r)}$, and let a dot denote differentiation of R_s with respect to s . At almost every $x \neq 0$,

$$DT_\theta(x) = Q(r) + \theta'(r)(\dot{R}_{\theta(r)}x) \otimes e_r. \quad (4.2)$$

Consequently $|DT_\theta| \leq C_n(1 + r|\theta'|)$, and (4.1), together with polar coordinates, gives $DT_\theta \in L^p(\mathbb{B}^n)$. Formula (4.2) is initially a local weak-derivative formula away from the origin. Integration by parts on $\mathbb{B}^n \setminus \overline{\mathbb{B}(0, \varepsilon)}$ produces an inner boundary term of size $O(\varepsilon^n)$ because $|T_\theta(x)| = \varepsilon$ there. It vanishes as $\varepsilon \downarrow 0$, proving that the displayed formula is the global weak derivative.

The determinant is exactly one. Since $S_s = R_s^T \dot{R}_s$ is skew-symmetric, the matrix determinant lemma and (4.2) give

$$\begin{aligned} \det DT_\theta &= \det Q(r) \left(1 + e_r^T Q(r)^T \theta'(r) \dot{R}_{\theta(r)} x \right) \\ &= 1 + \frac{\theta'(r)}{r} x^T S_{\theta(r)} x = 1 \end{aligned}$$

almost everywhere.

The identity $|T_\theta(x)| = |x|$ gives a global inverse,

$$T_\theta^{-1}(y) = R_{-\theta(|y|)}y.$$

Both maps are continuous away from zero and are continuous at zero because their norm equals the radius. Thus T_θ is a homeomorphism of the ball onto itself.

It remains to construct the approximation. For $0 < \delta < 1/2$, freeze the angle at radius δ :

$$\theta^\delta(r) = \begin{cases} \theta(\delta), & 0 < r < \delta, \\ \theta(r), & \delta \leq r \leq 1. \end{cases}$$

The maps differ only in the ball of radius δ . From (4.2), using boundedness of rotations and their first derivatives,

$$\|T_{\theta^\delta} - T_\theta\|_{W^{1,p}(\mathbb{B}^n)}^p \leq C_{n,p} \left(\delta^n + \int_0^\delta r^{n-1+p} |\theta'(r)|^p dr \right), \quad (4.3)$$

which tends to zero. Also $\|T_{\theta^\delta} - T_\theta\|_\infty \leq 2\delta$.

For fixed δ , the function θ^δ belongs to $W^{1,p}(0,1)$ and is constant near zero. Extend it constantly to the left, mollify with radius smaller than $\delta/4$, and, if necessary, add a smooth correction supported near $r = 1$ to recover the endpoint value $\theta(1)$. This produces smooth η_m , constant near zero and with $\eta_m(1) = \theta(1)$, such that

$$\|\eta_m - \theta^\delta\|_{W^{1,p}(0,1)} + \|\eta_m - \theta^\delta\|_\infty \longrightarrow 0.$$

The derivative formula gives the pointwise estimate

$$|DT_{\eta_m} - DT_{\theta^\delta}| \leq C_n \left(|\eta_m - \theta^\delta| + r|\eta'_m - (\theta^\delta)'| + r|(\eta_m - \theta^\delta)'| |\eta_m - \theta^\delta| \right).$$

Hence $T_{\eta_m} \rightarrow T_{\theta^\delta}$ in $W^{1,p}$ and uniformly. Taking a diagonal sequence with $\delta \downarrow 0$ and using (4.3) yields the required θ_k .

Because every θ_k is constant near zero, T_{θ_k} is smooth at zero. The determinant calculation is pointwise, and the explicit inverse is smooth, so it is an incompressible diffeomorphism of the ball onto itself. Finally $\theta_k(1) = \theta(1)$ gives equality on the boundary sphere. $\square$

Remark 5 (A low-regularity example). The theorem permits arbitrarily many turns near the origin. For example, $\theta(r) = \log r$ has no limit as $r \downarrow 0$, while $r\theta'(r) = 1$ and condition (4.1) holds for every finite p . The resulting homeomorphism is still uniformly and strongly Sobolev approximable by exact incompressible diffeomorphisms of the ball.

5 Verification and scope

The proof uses no unproved analytic or computational dependency. The accompanying file `verification.md` gives an adversarial step check. The symbolic script `code/check_jacobians.py` verifies representative three- and four-dimensional unitriangular examples and two- and three-dimensional polynomial-angle radial twists. Those computations are sanity checks only; the proof is the triangular and rank-one determinant calculation above.

The triangular result does not generally keep the image $F(Q)$ fixed while smoothing; its approximants are diffeomorphisms onto their own images. The radial-twist result does keep both domain and image equal to $\mathbb{B}^n$, and it preserves the boundary trace. Neither result handles an arbitrary incompressible Sobolev homeomorphism, so Open Problem 5 remains open in general.

Bounded novelty check

On 9 August 2026 we searched the run’s registry, solution, attempt, and proof-gap indexes for arXiv:2502.01336 and for the terms “incompressible”, “volume preserving”, “triangular”, “laminate”, “shear”, and “radial twist”. We also searched arXiv/web indexes for Sobolev approximation by volume-preserving diffeomorphisms and exact determinant-one smoothing of shears or laminates. No exact match for either theorem above was found.

Two nearby results delimit the claim. Mueller proves uniform approximation results for volume-preserving homeomorphisms in a high-dimensional manifold setting, but the cited abstract does not assert strong $W^{1,p}$ convergence [2]. Campbell, D’Onofrio, and Vitek prove strong Sobolev diffeomorphic approximation of general locally finite piecewise affine homeomorphisms in dimensions three and four, including simultaneous inverse control, but their theorem does not assert exact incompressibility [3]. The present result is therefore recorded only as a candidate new structured partial theorem. The search was bounded rather than exhaustive, and the mechanism may be folklore.

Human-review recommendation

Review as a likely valid partial result. The most useful checks are:

1. confirm the simultaneous uniform/ $W^{1,p}$ density lemma at $p = 1$;
2. verify the distributional derivative across the origin in Theorem 4;
3. verify that the fixed-ball and boundary assertions use only radius preservation and the endpoint condition;
4. assess whether exact determinant-preserving triangular or radial-twist smoothing is already recorded in the specialist literature.

References

- [1] S. Hencl, *Ball–Evans approximation problem: recent progress and open problems*, arXiv:2502.01336v1, 2025.
- [2] S. Mueller, *Uniform approximation of homeomorphisms by diffeomorphisms*, arXiv:0901.1002, 2009.
- [3] D. Campbell, L. D’Onofrio, and T. Vitek, *Diffeomorphic approximation of piecewise affine homeomorphisms*, arXiv:2507.02854, 2025; *Journal of Geometric Analysis* 36 (2026), Article 157.